CS 451/551 Midterm 1

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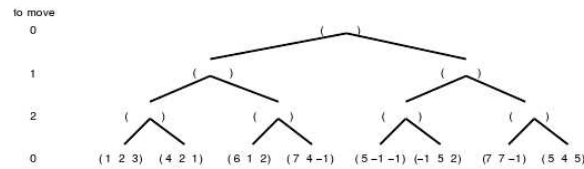
I. Question

Decide if each of the following is true or false. Explain why?

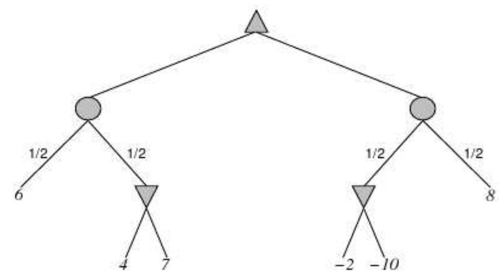
(a) Breadth-first is an optimal search algorithm.
(b) Simple reflex agents cope well with inaccessible environments.
(c) Minimax and alpha-beta can sometimes return different results.
(d) It is possible to write an exact evaluation function for chess.
(e) Assume that a king can move one square in any direction on a chessboard (8 directions in all). Manhattan distance is then an admissible heuristic for the problem of moving the king from square A to square B.
(f) It is possible to build a knowledge-based agent that is a pure reflex agent.
(g) A rational agent outperforms all nonrational agents because it knows the actual outcome of its actions.
(h) Simple hill-climbing is a complete algorithm for solving CSPs.
(i) h(n) = 0 is an admissible heuristic for 8-puzzle.
(j) Some pruning is possible in game trees with chance nodes.
(k) A perfectly rational backgammon agent never loses.
(l) Search algorithms cannot be applied in completely unobservable environments.

II. Question

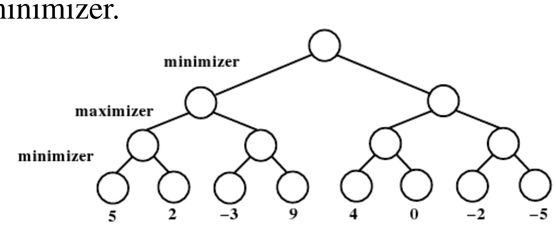
(a) Draw the smallest possible game tree on which alpha-beta will prune at least one leaf node. Make sure to label the leaves with values, and circle the leaf (or leaves) that will be pruned.
(b) Copy and complete the following game tree by filling in the backed-up value triples for all remaining nodes:



(c) Copy and complete the following game tree by filling in the values for all remaining nodes:



(d) Apply the minimax algorithm to the game tree below, where it is the minimizer's turn to play. Report the estimated values of the intermediate nodes and indicate the proper move of the minimizer.



Indicate, by crossing out, one (1) unnecessary call to the static board evaluator. Explain why this call to the board evaluator is unnecessary.

III. Question

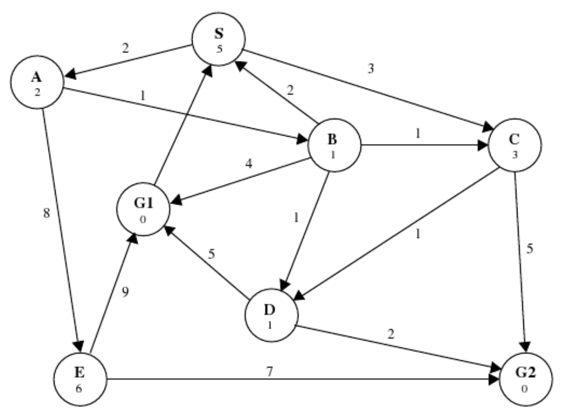
Consider the problem of placing k queens on an n × n chessboard such that no two queens are attacking each other, where k is given and $k \leq n^2$.

(a) Choose a CSP formulation. In your formulation, what are the variables?
(b) What are the values of each variable?
(c) What sets of variables are constrained, and how?
(d) Now consider the problem of putting as many queens as possible on the board without any attacks. We will solve this using local search.

Briefly describe in English a sensible successor function.

IV. Question

Consider the search space below, where S is the start node and G1 and G2 satisfy the goal test. Arcs are labeled with the cost of traversing them and the estimated cost to a goal is reported inside nodes. For each of the following search strategies, indicate which goal state is reached (if any) and list, in order, all the states popped off of the OPEN list. When all else is equal, nodes should be removed from OPEN in alphabetical order.



(a) Iterative Deepening
(b) A^*
(c) Hill climbing
(d) Breadth-first
(e) Depth-first

V. Question

Consider the following fitness function:

$$fitness = 5a + 3bc – d + 2e \tag{1}$$

where a-e are all Boolean-valued parameters. Compute the fitness of each of the members of the initial population below. Also compute the probability that each member of the population will be selected during the fitness-proportional reproduction process.

a	b	c	d	e
1	1	0	1	1
0	1	1	0	1
1	1	0	0	0
1	0	1	1	1
1	0	0	0	0

Assuming the first two of members of the population are selected for reproduction, and the crossover point is that between the b and the c, show the resulting children.

VI. Question

Represent the following sentences in predicate calculus.

(a) All PCs are computers.
(b) If someone owns a PC then there is some computer that they own.
(c) MaryBeth owns a PC.
(d) Anyone who owns a computer is a dweeb.

Then, show exactly how backward chaining solves MaryBeth is a dweeb.

VII. Question

(a) Given the interpretation P = true, Q = false, determine the truth value of the following

$$[(P \wedge Q) \Rightarrow Q] \Leftrightarrow [P \vee \neg Q]$$

(b) Is the following valid? Justify your answer using a truth table.

$$[(P \vee Q) \wedge (Q \vee R)] \Rightarrow (P \vee R)$$

c) A propositional 2-CNF expression is a conjunction of clauses, each containing exactly 2 literals.

$$(A \vee B) \wedge (\neg A \vee C) \wedge (\neg B \vee D) \wedge (\neg C \vee G) \wedge (\neg D \vee G)$$

Prove using resolution that the above sentence entails G.